CHAPTER 5 CIRCULAR MOTION

Key idea: a resultant force towards the centre keeps changing the direction of the velocity.

1 ANGULAR QUANTITIES

$$\theta = \frac{s}{r} \qquad \omega = \frac{d\theta}{dt}$$

$$\omega = \frac{2\pi}{T} = 2\pi f \qquad v = r\omega$$

1 radian: the angle subtended at the centre by an arc equal in length to the radius.

$360° = 2\pi$ rad; 1 rev $s^{-1} = 2\pi$ rad s^{-1}

2 CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r} = r\omega^2$$

- Directed towards the **centre**
- Speed constant, velocity changes direction, so the body accelerates

3 CENTRIPETAL FORCE

$$F = \frac{mv^2}{r} = mr\omega^2$$

- Not a new force: provided by tension, friction, gravity or normal reaction
- Perpendicular to v: does **no work**; E_k constant
- Removed: body moves off along the **tangent**

4 HORIZONTAL CIRCLES

Level bend (friction provides the force):

$$\mu mg \geq \frac{mv^2}{r} \qquad v_{\max} = \sqrt{\mu g r}$$

Banked track, no friction:

$$\tan\theta = \frac{v^2}{rg}$$

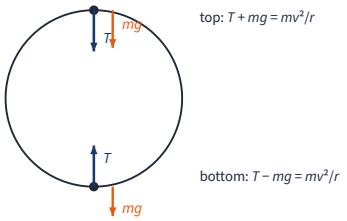
Conical pendulum:

$$T\cos\theta = mg$$

$$T\sin\theta = \frac{mv^2}{r}$$

Resolve vertically (balance) and horizontally (centripetal).

5 VERTICAL CIRCLE

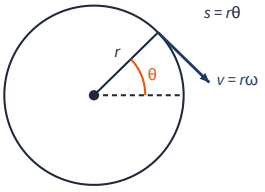


$$\text{Top: } T + mg = \frac{mv^2}{r}$$

$$\text{Bottom: } T - mg = \frac{mv^2}{r}$$

- Tension is greatest at the **bottom**
- Just completes the loop: $T = 0$ at top, $v_{\text{top}} = \sqrt{gr}$
- Convex bridge top: $mg - N = \frac{mv^2}{r}$

6 UNIFORM CIRCULAR MOTION



Quantity	Constant?
speed, ω, E_k	yes
velocity	no (direction)
acceleration	no (direction)

7 UNIT CHECK

Quantity	Formula	Unit
Angular displacement	$\theta = s/r$	rad
Angular velocity	$\omega = 2\pi f$	rad s^{-1}
Centripetal acceleration	v^2/r	m s^{-2}
Centripetal force	$mr\omega^2$	N

8 COMMON SECTION A TRAPS

- Adding a "centrifugal force" to the free-body diagram.
- Using degrees instead of radians in $s = r\theta$.
- Saying velocity is constant (only speed is).
- Using the diameter instead of the radius.
- At the top, forgetting that T and mg both point down.

EXAM TIPS

- ✓Draw the forces, then mark the centre of the circle.
- ✓Net force towards centre $= mv^2/r$.
- ✓Use energy conservation to link top and bottom speeds.
- ✓Convert rev min^{-1} to rad s^{-1} first.